



CMPG325 NETWORKS PROJECT

Milestone 2



AMUKELANI MANYAGWE
47932945

1. NETWORK IMPLEMENTATION

Working Packet Tracer Implementation

The proposed network design from Milestone 1 was implemented using Cisco Packet Tracer. The implementation translated the physical and logical network designs into a functional network environment consisting of a router, two switches, end-user devices and a server.

The network was divided into separate VLANs according to the functional requirements identified during the design phase. VLAN segmentation was implemented for the Administration, Operations, Server, Future Floor and Management networks. Router-on-a-stick inter-VLAN routing was implemented on R1 to allow controlled communication between the different network segments.

The implementation was subsequently tested using connectivity tests and device verification commands to ensure that the configured network operated according to the proposed design.

Network Components

Component	Role
R1	Inter-VLAN routing and network gateway
SW1-MAIN	Main access/distribution switch
SW2-FUTURE	Future Floor switch
SRV1	Server on VLAN 30
Administration PCs	For Users on VLAN 10
Operations PCs	For Users on VLAN 20
Future Floor PCs	For Users on VLAN 40
AccessPoint-PT	Wireless access on the Future Floor

Network Addressing

Network	Subnet	Gateway
VLAN 10 – ADMIN	192.168.28.0/27	192.168.28.1
VLAN 20 – OPERATIONS	192.168.28.32/27	192.168.28.33
VLAN 30 – SERVER	192.168.28.64/28	192.168.28.65
VLAN 40 – FUTURE	192.168.28.96/27	192.168.28.97
VLAN 99 – MANAGEMENT	192.168.28.128/28	192.168.28.129

Physical Implementation

The physical topology was implemented using R1, SW1-MAIN and SW2-FUTURE. SW1-MAIN acts as the primary switch for the Administration, Operations and Server segments, while SW2-FUTURE represents the switching infrastructure for the Future Floor. The two switches are interconnected using an 802.1Q trunk, allowing multiple VLANs to be transported between the switching infrastructure.

The server is connected directly to SW1-MAIN and is assigned to VLAN 30. Administration devices are connected to VLAN 10, while Operations devices are connected to VLAN 20. Future Floor devices, including two representative PCs and an AccessPoint-PT, are connected to SW2-FUTURE and assigned to VLAN 40.

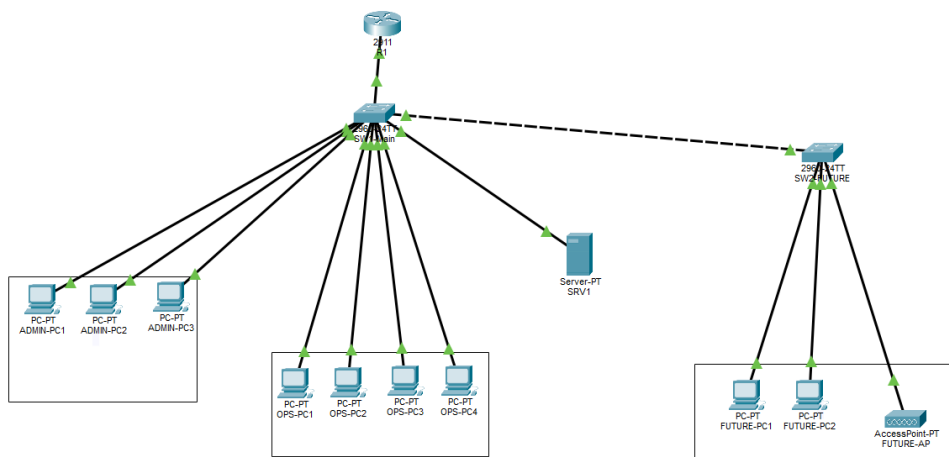


Figure 1: Implemented logistics network topology in Cisco Packet Tracer

VLAN Implementation

VLAN segmentation was implemented to separate users and network resources according to their functional roles. Administration devices were assigned to VLAN 10, Operations devices to VLAN 20, the server to VLAN 30 and Future Floor devices to VLAN 40. VLAN 99 was reserved specifically for network-device management.

Access ports were assigned to their appropriate VLANs, while the inter-switch connection was configured as an 802.1Q trunk. This allowed the required VLANs to traverse the connection between SW1-MAIN and SW2-FUTURE.

VLAN verification

```
SW1-MAIN>enable
SW1-MAIN#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
10	ADMIN	active	Fa0/1, Fa0/2, Fa0/3
20	OPERATIONS	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7
30	SERVER	active	Fa0/8
40	FUTURE	active	
99	MANAGEMENT	active	
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdinet-default	active	
1005	trnet-default	active	

Figure 2: VLAN configuration and port assignments on SW1-MAIN.

```
SW2-FUTURE>enable
SW2-FUTURE#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10 ADMIN	active	
20 OPERATIONS	active	
30 SERVER	active	
40 FUTURE	active	Fa0/1, Fa0/2, Fa0/3
99 MANAGEMENT	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

Figure 3: VLAN configuration and port assignments on SW2-future.

Trunk Implementation

An IEEE 802.1Q trunk was configured between SW1-MAIN and SW2-FUTURE. The trunk carries VLANs 10, 20, 30, 40 and 99 between the switches. This configuration allows devices connected to the Future Floor switch to communicate with the appropriate network gateways through the main switching infrastructure.

Trunk verification

```
SW1-MAIN#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	1
Gig0/2	on	802.1q	trunking	1


```
Port Vlans allowed on trunk
Gig0/1 1-1005
Gig0/2 1-1005
```



```
Port Vlans allowed and active in management domain
Gig0/1 1,10,20,30,40,99
Gig0/2 1,10,20,30,40,99
```



```
Port Vlans in spanning tree forwarding state and not pruned
Gig0/1 1,10,20,30,40,99
Gig0/2 1,10,20,30,40,99
```

Figure 4: Verification of the 802.1Q trunk between the switching infrastructure SW1.

```
SW2-FUTURE#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/4	on	802.1q	trunking	1


```
Port Vlans allowed on trunk
Fa0/4 1-1005
```



```
Port Vlans allowed and active in management domain
Fa0/4 1,10,20,30,40,99
```



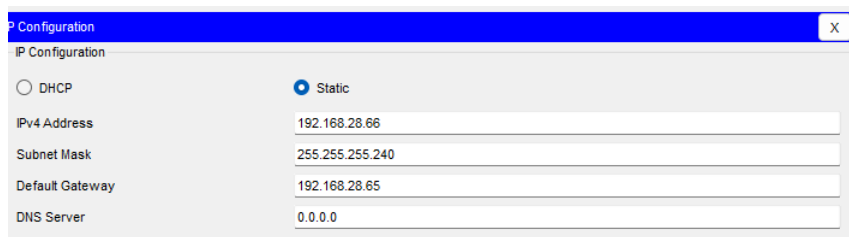
```
Port Vlans in spanning tree forwarding state and not pruned
Fa0/4 1,10,20,30,40,99
```

Figure 5: Verification of the 802.1Q trunk between the switching infrastructure SW2.

Server Implementation

SRV1 was connected directly to SW1-MAIN and assigned to VLAN 30. A static IP address was configured so that the server could provide a consistent destination for users and network services. The server was configured with IP address 192.168.28.66, subnet mask 255.255.255.240 and default gateway 192.168.28.65.

Server configuration



IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.28.66

Subnet Mask: 255.255.255.240

Default Gateway: 192.168.28.65

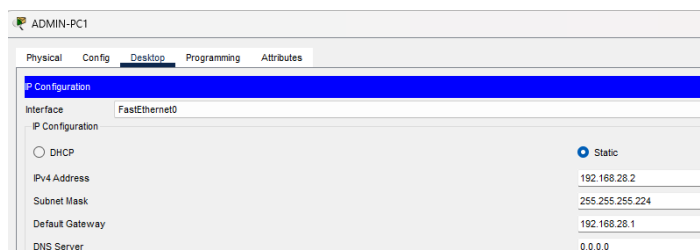
DNS Server: 0.0.0.0

Figure 6: Static IP configuration of SRV1 within the Server VLAN.

End-Device Implementation

Representative end-user devices were configured using static IPv4 addressing according to their respective VLANs. Administration PCs were assigned addresses from the VLAN 10 subnet, Operations PCs from VLAN 20 and Future Floor PCs from VLAN 40. The default gateway assigned to each device corresponds to the R1 sub interface for its VLAN.

Device group	Example IP	Gateway
ADMIN-PC1	192.168.28.2	192.168.28.1
OPS-PC1	192.168.28.34	192.168.28.33
FUTURE-PC1	192.168.28.98	192.168.28.97
SRV1	192.168.28.66	192.168.28.65



ADMIN-PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

☐ DHCP ☒ Static

IPv4 Address: 192.168.28.2

Subnet Mask: 255.255.255.224

Default Gateway: 192.168.28.1

DNS Server: 0.0.0.0

Figure 7: ADMIN-PC1 example.

Management VLAN

VLAN 99 was implemented as a dedicated Management VLAN for network infrastructure. Rather than assigning management addresses to the user VLANs, R1, SW1-MAIN and SW2-FUTURE were given management addresses within VLAN 99. This provides a separate network segment for administrative access to the network infrastructure.

```

SW1-MAIN>enable
SW1-MAIN#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	up	up
FastEthernet0/4	unassigned	YES	manual	up	up
FastEthernet0/5	unassigned	YES	manual	up	up
FastEthernet0/6	unassigned	YES	manual	up	up
FastEthernet0/7	unassigned	YES	manual	up	up
FastEthernet0/8	unassigned	YES	manual	up	up
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down
FastEthernet0/22	unassigned	YES	manual	down	down
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	manual	up	up
Vlan1	unassigned	YES	manual	administratively down	down
Vlan99	192.168.28.131	YES	manual	up	up

Figure 8: management interface SW1.

```

SW2-FUTURE#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	up	up
FastEthernet0/4	unassigned	YES	manual	up	up
FastEthernet0/5	unassigned	YES	manual	down	down
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down
FastEthernet0/22	unassigned	YES	manual	down	down
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	down	down
GigabitEthernet0/2	unassigned	YES	manual	down	down
Vlan1	unassigned	YES	manual	administratively down	down
Vlan99	192.168.28.132	YES	manual	up	up

Figure 9: management interface SW2.

Inter-VLAN Routing

Inter-VLAN routing was implemented using a router-on-a-stick configuration on R1. Separate sub interfaces were configured for VLANs 10, 20, 30, 40 and 99. Each sub interface provides the default gateway for its respective VLAN.

This configuration allows devices located in different VLANs to communicate where required. For example, Administration and Operations users can access resources located within the Server VLAN without placing all devices into the same broadcast domain.

Router subinterfaces

```

R1>enable
R1#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	unset	up	up
GigabitEthernet0/0.10	192.168.28.1	YES	manual	up	up
GigabitEthernet0/0.20	192.168.28.33	YES	manual	up	up
GigabitEthernet0/0.30	192.168.28.65	YES	manual	up	up
GigabitEthernet0/0.40	192.168.28.97	YES	manual	up	up
GigabitEthernet0/0.99	192.168.28.129	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

Figure 10: R1 subinterfaces providing gateways for the implemented VLANs.

```

interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.28.1 255.255.255.224
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.28.33 255.255.255.224
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.28.65 255.255.255.240
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.28.97 255.255.255.224
!
interface GigabitEthernet0/0.99
 encapsulation dot1Q 99
 ip address 192.168.28.129 255.255.255.240

```

Figure 11: R1 router-on-a-stick configuration

Device Configuration Evidence

The configuration of the primary network devices was verified using Cisco IOS commands. Configuration evidence was included for R1 and SW2-FUTURE to demonstrate that the routing, VLAN and management components of the network were implemented according to the proposed design.

R1 was verified to ensure that the required VLAN subinterfaces were configured and operational. These subinterfaces provide the default gateway for VLANs 10, 20, 30, 40 and 99 and enable inter-VLAN routing.

SW2-FUTURE was verified to confirm that the Future Floor VLAN was correctly implemented, that the required ports were assigned to VLAN 40 and that the trunk connection was operational. Its VLAN 99 management interface was also verified to support remote management through SSH.

2. ASSIGNED FEATURE & VERIFICATION

Assigned Feature: Secure Shell (SSH) Remote Management

Secure Shell (SSH) was implemented as the assigned network feature to provide secure remote management of the network infrastructure. The feature allows an authorised administrator to remotely access the router and switches through the network without requiring direct physical access to each device. SSH was implemented on R1, SW1-MAIN and SW2-FUTURE. The devices use the dedicated Management VLAN 99 for management traffic, separating network administration from normal user traffic.

Why SSH Was Implemented

Network devices such as routers and switches require administrative access for configuration, troubleshooting and maintenance. Direct console access requires an administrator to be physically present at the device, which is not always practical in a logistics environment where network equipment may be in dedicated network rooms.

SSH provides a method of remotely accessing the network infrastructure while using authentication and encrypted communication. In this implementation, SSH was therefore used to provide remote administrative access to R1, SW1-MAIN and SW2-FUTURE.

The use of VLAN 99 for management further separates administrative traffic from the normal user networks.

SSH Addressing

Device	Management Interface	Management IP	VLAN
R1	G0/0.99	192.168.28.129	99
SW1-MAIN	VLAN 99	192.168.28.131	99
SW2-FUTURE	VLAN 99	192.168.28.132	99

SSH Implementation

Secure Shell (SSH) was implemented consistently across **R1**, **SW1-MAIN** and **SW2-FUTURE** to provide secure remote management of the network infrastructure. The implementation was performed using the same general configuration process on each device, with only the hostname and management IP address differing between devices.

First, each device was given a unique hostname and the domain name `logistics.local` was configured. A local administrative user with privilege level 15 was then created to provide authenticated access. RSA cryptographic keys were generated on each device to enable the SSH service.

The VTY (Virtual Terminal) lines were subsequently configured to use the local user database for authentication and to accept **SSH connections only**. The command `transport input ssh` was used to prevent Telnet access through the VTY lines, ensuring that remote management was performed using SSH. Finally, the configurations were saved to the startup configuration so that they would remain available after a device restart.

SSH Verification

```
SW1-MAIN>enable
SW1-MAIN#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
```

Figure 12: SSH is enabled on the SW-1 device

```
R1#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
```

Figure 13: SSH is enabled on the R1 device

```
SW2-FUTURE#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
```

Figure 14: SSH is enabled on the SW-2 device

```
SW1-MAIN#show running-config | section line vty
line vty 0 4
  login local
  transport input ssh
line vty 5 15
  login local
  transport input ssh
```

Figure 15: VTY SW1

```
R1#show running-config | section line vty
line vty 0 4
  login local
  transport input ssh
line vty 5 15
  login local
  transport input ssh
```

Figure 16: VTY R1

```
SW2-FUTURE#show running-config | section line vty
line vty 0 4
  login local
  transport input ssh
line vty 5 15
  login local
  transport input ssh
```

Figure 17: VTY SW2

SSH Remote Access Testing

SW1-MAIN

```
C:\>ssh -l admin 192.168.28.131
Password:

SW1-MAIN>
SW1-MAIN>
```

Figure 18: Successful SSH connection from ADMIN-PC1 to SW1-MAIN.

SW2-FUTURE

```
C:\>ssh -l admin 192.168.28.132
Password:

SW2-FUTURE>enable
% No password set.
SW2-FUTURE>
```

Figure 19: Successful SSH connection from ADMIN-PC1 to SW2-FUTURE.

R1

```
C:\>ssh -l admin 192.168.28.129
Password:
R1#
R1#
```

Figure 20: Successful SSH connection from ADMIN-PC1 to R1.

Gateway Connectivity

The representative PCs were tested against their respective VLAN gateways.

Device	Destination	Result
ADMIN-PC1	192.168.28.1	Successful
OPS-PC1	192.168.28.33	Successful
FUTURE-PC1	192.168.28.97	Successful

Inter-VLAN Connectivity from the PCs

Inter-VLAN connectivity was tested by sending ICMP requests from representative devices in the Administration, Operations and Future Floor VLANs to the server located in VLAN 30. Successful responses demonstrate that R1 is correctly routing traffic between the different VLANs.

ADMIN-PC1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.28.1

Pinging 192.168.28.1 with 32 bytes of data:

Reply from 192.168.28.1: bytes=32 time<1ms TTL=255
Reply from 192.168.28.1: bytes=32 time<1ms TTL=255
Reply from 192.168.28.1: bytes=32 time<1ms TTL=255
Reply from 192.168.28.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.28.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.28.66

Pinging 192.168.28.66 with 32 bytes of data:

Reply from 192.168.28.66: bytes=32 time<1ms TTL=127
Reply from 192.168.28.66: bytes=32 time<1ms TTL=127
Reply from 192.168.28.66: bytes=32 time<1ms TTL=127
Reply from 192.168.28.66: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.28.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 20: Successful connectivity from ADMIN-PC1 to the Server VLAN.

OPERATIONS-PC1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.28.33

Pinging 192.168.28.33 with 32 bytes of data:

Reply from 192.168.28.33: bytes=32 time<1ms TTL=255
Reply from 192.168.28.33: bytes=32 time<1ms TTL=255
Reply from 192.168.28.33: bytes=32 time<1ms TTL=255
Reply from 192.168.28.33: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.28.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.28.66

Pinging 192.168.28.66 with 32 bytes of data:

Reply from 192.168.28.66: bytes=32 time<1ms TTL=127
Reply from 192.168.28.66: bytes=32 time<1ms TTL=127
Reply from 192.168.28.66: bytes=32 time<1ms TTL=127
Reply from 192.168.28.66: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.28.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 21: Successful connectivity from OPERATIONS-PC1 to the Server VLAN.

FUTURE-PC1

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.28.97

Pinging 192.168.28.97 with 32 bytes of data:

Reply from 192.168.28.97: bytes=32 time<1ms TTL=255
Reply from 192.168.28.97: bytes=32 time<1ms TTL=255
Reply from 192.168.28.97: bytes=32 time<1ms TTL=255
Reply from 192.168.28.97: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.28.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 22: Successful connectivity from FUTURE-PC1 to the Server VLAN.

Conclusion

The assigned SSH remote-management feature was successfully implemented across R1, SW1-MAIN and SW2-FUTURE. A dedicated Management VLAN was used to separate network-device management traffic from the user and server networks.

Testing confirmed that the network devices could be reached using SSH from an authorised workstation. Additional connectivity tests verified VLAN operation, trunking, gateway connectivity and inter-VLAN communication.

The successful implementation and testing demonstrate that the network design developed during Milestone 1 has been translated into a functional network environment with secure remote-management capabilities.

The network designed during Milestone 1 was successfully implemented in Cisco Packet Tracer. VLAN segmentation, inter-VLAN routing, trunking, server connectivity and network management were configured and tested. SSH was implemented as the assigned network feature, providing secure remote management of R1, SW1-MAIN and SW2-FUTURE through the dedicated Management VLAN. The completed Packet Tracer implementation, testing evidence and updated GitHub repository provide evidence of the completed Milestone 2 requirements.

Testing Summary

Test	Purpose	Result
VLAN verification	Confirm VLANs and port assignments	Pass
Trunk verification	Confirm inter-switch VLAN transport	Pass
ADMIN gateway test	Verify VLAN 10 gateway connectivity	Pass
OPERATIONS gateway test	Verify VLAN 20 gateway connectivity	Pass
FUTURE gateway test	Verify VLAN 40 gateway connectivity	Pass
ADMIN → Server	Verify inter-VLAN routing	Pass
OPERATIONS → Server	Verify inter-VLAN routing	Pass
FUTURE → Server	Verify inter-VLAN routing	Pass
Management connectivity	Verify VLAN 99 connectivity	Pass
SSH → R1	Verify remote management	Pass
SSH → SW1-MAIN	Verify remote management	Pass
SSH → SW2-FUTURE	Verify remote management	Pass

GitHub Portfolio

The GitHub repository was updated to provide a central location for the project documentation, network designs, Cisco Packet Tracer implementation and testing evidence. The repository demonstrates the progression of the project from the initial network design in Milestone 1 to the implemented and tested network in Milestone 2.

[Elbowg/CMPG235-Computer-Networks-Project](#)